Master thesis in Brain and Cognition

Investigating newborns' sensitivity to the Sonority Sequencing Principle (SSP) using functional Near-Infrared Spectroscopy (fNIRS)

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1 Abstract

Background: The Sonority Sequencing Principle (SSP) is a universal phonological constraint which describes the preferable organisation of sounds within syllables. Previous research by Gómez et al. (2014) demonstrated that newborns discriminate between syllables that follow or violate the SSP, suggesting that sensitivity to sonority-based constraints may be present from birth.

Research question: The present study aimed to conceptually replicate the findings of Gómez et al. (2014) by investigating whether newborns discriminate between well-formed (SSP-following) and ill-formed (SSP-violating) syllables using functional near-infrared spectroscopy (fNIRS).

Methods: Data were collected from 33 healthy full-term newborns while they listened to SSP-following and SSP-violating syllables during sleep. Haemodynamic responses were preprocessed and channels were classified into region-of-interest groupings. Mean HbO and HbR concentration changes were compared between conditions using one-sampled t-tests.

Results: Significantly greater HbO responses to SSP-following compared to SSP-violating syllables were observed in the left frontal, left temporal, and right temporal regions. No significant differences were found for HbR. Overall, the findings conceptually replicated the principal conclusion of Gómez et al. (2014), nevertheless with differences in spatial distribution of significant responses between the two studies.

Conclusions: The findings provide further evidence that newborns discriminate between well-formed and ill-formed syllables from the first days of life, supporting Gómez et al. (2014) finding that sensitivity to sonority-based constraints is present at birth. Furthermore, the study establishes a foundation for a subsequent study investigating whether this sensitivity reflects language-specific phonological processing or more general auditory processing mechanisms using non-linguistic tone stimuli.

Keywords: infant cognition; functional near-infrared spectroscopy (fNIRS); Sonority Sequencing Principle; early syllable processing; conceptual replication.

2 Introduction

Although perceived like a continuous stream of sounds, speech consists of multiple hierarchical linguistic units such as phrases made of words, words made of syllables, and syllables made of phonemes (Meyer, 2017). The ability to segment this continuous speech stream into smaller units is a critical mechanism of language acquisition. While this process appears effortless for adults, given their extensive experience with language, it is remarkable that newborn infants already show the ability to parse the speech into syllable-sized units (Flo et al 2022). Understanding the mechanisms that underlie such an early segmentation ability is therefore central to explaining what drives infants' rapid acquisition of language.

This thesis aims to investigate whether early sensitivity to the syllable may be related to a universal linguistic constraint on the internal structure of syllables known as the Sonority Sequencing Principle (SSP). The following sections first review evidence on early language processing in infancy, focusing on the syllable as a perceptually salient unit during early language acquisition. The Sonority Sequencing Principle (SSP) is then introduced as a universal constraint determining the structure of syllables across languages. Next, functional Near-Infrared Spectroscopy (fNIRS) is described as the neuroimaging method used in this study. Finally, the current study, goals, and research questions are presented.

2.1 Early language processing

From birth onwards, infants begin to identify units in speech including phonemes, syllables, and words, which serve as the fundamental building blocks for subsequent linguistic processing (Friederici, 2005; Werker, 2018; Saffran & Kirkham, 2018).

Importantly, language processing may also begin even before birth. The intrauterine environment is unique, with constant, primarily low-frequency sounds of mother's cardiovascular and digestive systems and voice transmitted to the fetal ear via amniotic fluid (Gerhardt & Abrams, 2000). Evidence on prenatal speech processing shows that fetuses from 28-30 gestational weeks are able to extract prosodic information of the languages heard in utero (Gervain, 2019). As a result, this prenatal experience speech already shapes infants' perception abilities.

In addition, research shows that, at birth, newborns can discriminate between syllables. Specifically, Bijeljac-Babic et al. (1993) showed that four-day old newborns were able to discriminate multisyllabic utterances that differed in the number of syllables but failed to discriminate stimuli that differed only in the number of phonemes. This suggests that newborns are sensitive to syllable-level structure which points to the idea that syllables constitute a perceptually salient unit in early speech perception. Furthermore, neurophysiological evidence further supports the relevance of syllables as a perceptually salient linguistic unit by showing that newborns preferentially parse speech into units that correspond to syllables. Flo et al. (2022) showed that the newborn brain spontaneously synchronizes with speech at the frequency rate of the syllable (5-7 Hz), also found in adults (Poeppel, 2003). In addition, such privileged processing of syllables seems to be in place in extremely pre-term neonates. A study by Mahmoudzadeh et al. (2012) found that pre-term newborns of 28 and 32 gestational age discriminated between syllables such as /ba/ and /ga/. This suggested that mechanisms supporting syllable discrimination are functional at the onset of cortical organisation for auditory perception (~28 gestational weeks). Taken together, this body of evidence suggests that the syllable may constitute one of the earliest and most salient units at the very onset of language acquisition.

2.2 The Sonority Sequencing Principle (SSP)

Since syllables provide an entry point to language learning, it becomes crucial to understand the rules defining the structure of syllables. In Linguistics, the organisation of sounds within syllables is constrained by *universal principles* that govern the ordering of phonemes across languages (Clements, 1990; Selkirk, 1984; Parker, 2008). Such principles have been argued to account for both the cross-linguistic regularities in syllable structure and the special role of syllables as a fundamental unit in early language development (Friederici, 2005).

The main universal principle that determines the structure of syllables across most languages is the Sonority Sequencing Principle (SSP). Sonority refers to the intensity of a phoneme, reflecting differences in acoustic energy and articulatory openness (Parker, 2008, 2011). Vowels are the most sonorous phonemes as they carry high acoustic energy, whilst consonants such as stops (e.g., /p/, /t/) are among the least

sonorous as they carry a lower acoustic energy. In between we find intermediate classes of phonemes such as fricatives (e.g., /f/, /s/), nasals (e.g., /m/, /n/), and liquids (e.g., /l/, /r/). The SSP determines the ordering of phonemes within a syllable based on their intensity (Parker, 2008). The SSP states that phonemes should appear such that intensity increases from the onset of the syllable (the initial consonant or consonant cluster) to the nucleus (usually a vowel), peaks at the nucleus and decreases thereafter towards the coda (the final consonant; Clements, 1990; Selkirk, 1984; Parker, 2008). For example, as seen in Figure 1, in the syllable *blif*, sonority rises from /b/ to /l/ and reaches its peak at the vowel /i/, before decreasing towards the final consonant /f/. In contrast, in *lbif*, there is a drop in sonority before the vowel, as /b/ is less sonorous than /l/, thereby violating the SSP.

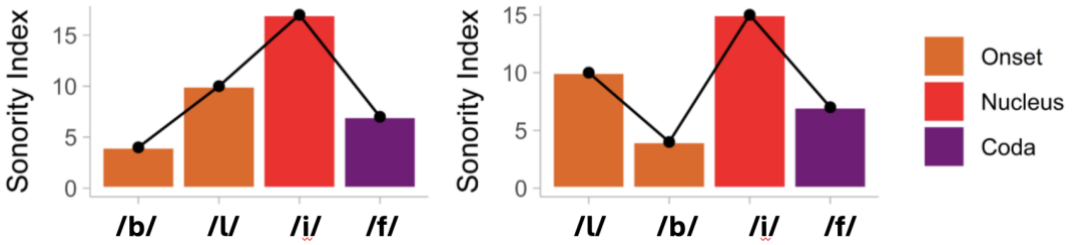


Figure 1. The Sonority Sequencing Principle in well-formed (*blif*) and ill-formed (*lbif*) syllables

Research using computational models trained on speech corpora with different languages has demonstrated that fluctuations in sonority are informative of syllable boundaries and can serve as reliable cues for segmenting continuous speech into syllable-sized units (Räsänen et al., 2018). This was achieved by the model without requiring prior knowledge of any specific language, supporting the assumption that syllable-like structure is accessible to infants from the earliest stages of language acquisition.

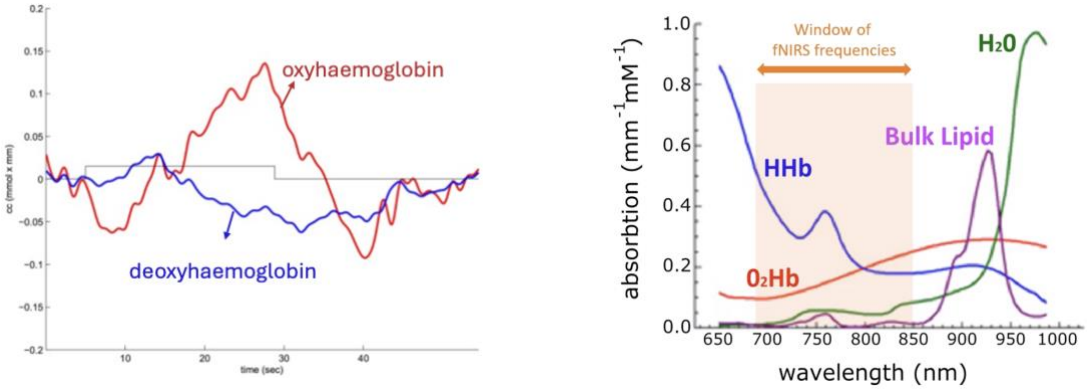
Sensitivity to sonority-based constraints has been observed in newborns. A study by Gómez et al. (2014) investigated whether two-day-old newborns are sensitive to violations of the SSP. The authors used functional Near-Infrared Spectroscopy (fNIRS) to examine differences in newborns' haemodynamic responses when presented with well-formed and ill-formed syllables. fNIRS measures relative changes in concentrations of oxyhaemoglobin (HbO) and deoxyhaemoglobin (HbR) in the cerebral blood, providing an indirect index of neural activity via the haemodynamic response (see Section 2.3 for a detailed explanation of fNIRS). The stimuli used were syllables

with a CCVC structure (C=consonants, V=vowels) that displayed either a sonority rise (well-formed) or a sonority fall (ill-formed) in the initial consonant cluster (e.g., ***bl****if* vs ***lb****if*). Results from 24 2-day-old newborns showed that haemodynamic responses in the left temporal cortex differed significantly between well-formed and ill-formed syllables. Well-formed syllables elicited a greater decrease in oxyhaemoglobin (HbO) compared to ill-formed syllables, reflecting an inverted haemodynamic response, commonly reported in newborn fNIRS studies (Issard & Gervain, 2018). The different haemodynamic responses across conditions suggested that newborns are sensitive to violations of the SSP.

2.3 Functional Near-Infrared Spectroscopy (fNIRS)

The study presented in this thesis has been conducted with functional Near-Infrared Spectroscopy (fNIRS). This is an optical neuroimaging method that provides a proxy of neural activity by measuring relative changes in oxygenated haemoglobin (HbO) and deoxygenated haemoglobin (HbR) in the brain (Villringer & Chance, 1997). fNIRS has been widely used in research with newborns because it is well-suited to studying speech perception in newborns (Gervain et al., 2008; Gómez et al., 2014; Marino et al., 2025). It can be used while newborns are sleeping, it is non-invasive, and relatively tolerant to small movements, making it suitable for this population. fNIRS relies on the mechanism of neurovascular coupling, whereby increased neural activity leads to a rise in metabolic demand, triggering an increase in cerebral blood flow (Logothetis et al., 2001). This typically results in an increase in HbO and a corresponding decrease in HbR in the activated brain regions. The pattern of variations in HbO and HbR concentrations over time defines the haemodynamic response function (HRF), which reflects how haemoglobin (Hb) concentrations evolve following neural activation (Issard & Gervain, 2022). The haemodynamic response in newborns is slower compared to adults at reaching the peak but also at returning to baseline (Meek et al., 1998). A canonical HRF is typically characterised by an initial increase in HbO alongside a decrease in HbR (see Figure 2A), however, studies with newborns and infants have also reported inverted haemodynamic responses, in which a decrease in HbO and an increase in HbR is observed (e.g., Abboub et al., 2016; Benavides-Varela et al., 2012; Nallet et al., 2023; Wagner et al., 2011).

Such inverted responses are not uncommon in infant studies. In particular, the balance between oxygen supply and demand – which determines the HRF – may differ in infants compared to adults (Lloyd-Fox et al., 2010). Immature neurovascular coupling in infants may result in insufficient blood flow to meet metabolic demand, producing an increase in HbR and a decrease in HbO, contrary to haemodynamic responses reported in adult studies. Consistent with this developmental explanation, a review by Issard and Gervain (2018) pointed that the haemodynamic response becomes increasingly canonical with age. Additionally, factors such as stimulus complexity, sensory modality, measured cortical region, and the infants' arousal state (e.g., sleep vs. wakefulness) have been suggested to modulate the haemodynamic response. Nevertheless, despite these proposed explanations, there is currently no consensus on the precise interpretation of the inverted haemodynamic response. As most fNIRS systems used for research only measure *relative* changes in HbO and HbR concentrations, this makes it difficult to infer a physiological and functional interpretation of these relative changes in oxygenation (Issard & Gervain, 2018).



(A) **(B)**

Figure 2. (A) Canonical HRF in a newborn (Issard & Gervain, 2018) **(B)** Differential light absorption spectra of HbO and HbR in NIR range (Kumar et al., 2017)

At the physical level, fNIRS relies on the differential near-infrared light absorption properties of HbO (absorbs NIR light mostly at 850nm) and HbR (absorbs NIR light mostly at 760nm) to detect the changes in concentration of both molecules (see Figure 2B). In continuous-wave (CW) systems, light is emitted continuously to measure the relative changes in light intensity, and thus the relative changes in HbO and HbR concentrations are estimated. Specifically, light is emitted through a source at two wavelengths (760nm and 850nm), travels through biological tissue, where it is absorbed

and scattered before reaching the detector. According to how much light of either wavelength reaches the detector, the relative changes in concentration of HbO and HbR are estimated. Each source-detector pair forms a channel which corresponds to a different region of the brain and the source-detector distance determines the depth of light penetration into the cortex.

2.4 The Current Study

The present thesis attempts to do a conceptual replication of Gómez et al. (2014) as they are the first and only study to report sensitivity to the SSP in humans at birth. The current aim is therefore to examine whether newborns can discriminate between well-formed (SSP-following) and ill-formed (SSP-violating) speech stimuli by measuring newborns' haemodynamic responses using fNIRS. Importantly, it is essential to establish the reliability of the finding that particularly because the subsequent experiment of the wider project will employ the same experimental design and fNIRS but with different, non-linguistic stimuli.

Despite the evidence by Gómez et al. (2014), it remains unclear whether sensitivity to the SSP at birth reflects a linguistic processing or a more general auditory processing. From a purely acoustic perspective, syllables are characterised by an arch-shaped structure with an initial rise of intensity and a subsequent decrease. Such arch-shaped signal is also present in the vocalizations of other species (e.g., birdsong; Mol et al., 2017; Tierny et al., 2011). As syllabic structure is determined by the SSP which closely mirrors these acoustic intensity patterns (arch-shaped envelope), this raises the possibility that newborns' sensitivity to sonority constraints may arise from domain-general processing of acoustic envelope structure rather than from a language-specific phonological mechanism.

To test this, the wider project will examine whether newborns are also sensitive to sonority constraints in non-linguistic stimuli, such as tones that mimic the acoustic envelope of syllables. Using non-linguistic analogues of syllables would allow for the isolation of the contribution of acoustic structure independently of linguistic content. It is hypothesised that if newborns are sensitive to sonority constraints in tone strings, this could suggest that this sensitivity is based on general acoustic processing rather than language-specific mechanisms. Alternatively, if sensitivity to sonority constraints is observed only for speech stimuli and not for non-linguistic stimuli, this would provide

evidence for an early language-specific mechanism underlying the early ability to identify syllables.

The wider project therefore comprises of two experiments: a first conceptual replication of Gómez et al. (2014) using the same stimuli, methods, and design and a second experiment applying this paradigm to non-linguistic stimuli using the same experimental design and method. Due to time constraints, the present thesis focuses only on the first experiment.

3 Methods

3.1 Participants

Newborns were recruited and tested at the Hospital Sant Joan de Déu in Barcelona (Spain). The inclusion criteria were being healthy and full-term, having a birth weight of at least 2700g, a minimum of 37 gestational weeks, an Apgar score of at least 8 at the 10th minute after delivery, and no hematologic, cardiologic or metabolic complications.

Forty-eight newborns were recruited and tested. Of these, 15 newborns were excluded from the final sample due to data not meeting the inclusion criterion (set prior to data collection) of at least two artifact-free blocks per condition. Therefore, 33 healthy, full-term newborns (18 females; age: 25.2±12.6 hours) contributed data to the final analysis (attrition rate of 31.25%). Sample size was determined with an a priori power analysis with a significance threshold of $\alpha=0.05$, and desired statistical power of $1-\beta=0.80$. The expected effect size were based on the smallest statistically significant effect size of the results by Gómez et al. (2014) which is 0.563. We used G*Power to run the power analysis (Faul et al., 2009). This resulted to the minimum sample size of 33 participants. Participants' head circumference ranged from 31.5 and 38 cm, with average Apgar scores of 10±0 at the tenth minute, an average gestational age of 39.5±1 weeks, and an average birth weight of 3472±306 grams. In addition, a table of the linguistic profiles of the families the newborns were born into can be found in Appendix A. This information is descriptive as there are no hypothesis about the linguistic profile of participants.

3.2 Stimuli

The stimuli were identical to those used in Gómez et al. (2014). Specifically, 8 SSP-following syllables and 8 SSP-violating syllables with a CCVC (C=consonant; V=vowel) structure were used. Table 1 presents the full list of syllables used. For example, SSP-following syllables included “bran” and “from” while SSP-violating syllables included “lbif” and “lvug”. As stated in Gómez et al. (2014), stimuli were recorded by a female native Russian speaker as Russian allows for all syllable types used in the experiment. The syllable sets were equivalent in terms of average pitch, duration, or number of feature changes between the two first consonants. Sound intensity was set to 70 dB for all syllables.

Table 1

List of speech stimuli used, as in Gómez et al. (2014).

SSP-following syllables	SSP-violating syllables
pras	rvug
fros	rvem
vlug	lbug
vlin	rveg
brud	rdos
vros	rfug
from	lvug
bran	lbif

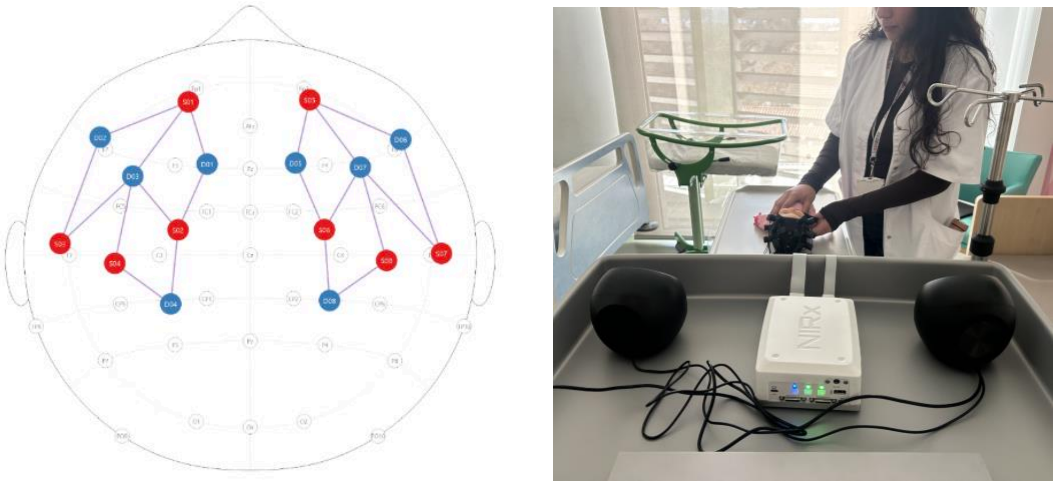
3.3 fNIRS acquisition system

Haemodynamic responses were recorded using the NIRSport2 system (NIRx, Berlin, Germany). This fNIRS machine emits continuous near-infrared light of two wavelengths (760 and 850 nm) to record the NIRS signal and was set at a sampling rate of 10.2 Hz, that is, samples were measured every 0.10 s. Sixteen optodes, 8 sources and 8 detectors were inserted into a stretchy cap (Easycap, Herrsching, Germany) of various head circumference sizes (32 cm to 38 cm). The optodes were arranged in a bilateral

chevron configuration, resulting in 20 channels with an inter-optode distance of 3 cm (see Figure 3A for the topographic layout of probe placement). The probe configuration covered the frontal, temporal, and parietal regions of both hemispheres and was compatible with the data-driven localisation analysis reported by Abboub et al. (2016), which was subsequently used to classify channels into anatomically defined regions of interest. The system was connected wirelessly to a computer, where the Aurora software (Version 23.9; NIRx, Berlin, Germany) allowed for fNIRS signal acquisition and visualisation in real time. The fNIRS system, computer, and loudspeakers were placed on a portable cart, allowing the experimental setup to be positioned next to the newborn's crib during testing (see Figure 3B). During each session, one experimenter controlled stimulus presentation and monitored signal quality, while a second experimenter attended to the newborn, monitoring their behavioural state, repositioning the cap if necessary, and providing comfort to the newborn when needed.

Figure 3

Experimental setup used for fNIRS data acquisition.



(A) **(B)**

Note. (A) Probe layout showing the arrangement of eight sources (red) and eight detectors (blue) forming 20 recording channels. **(B)** Experimental setup in the maternity ward, showing the portable hospital cart housing the NIRSport2 system, computer, and loudspeakers positioned beside the newborn's crib.

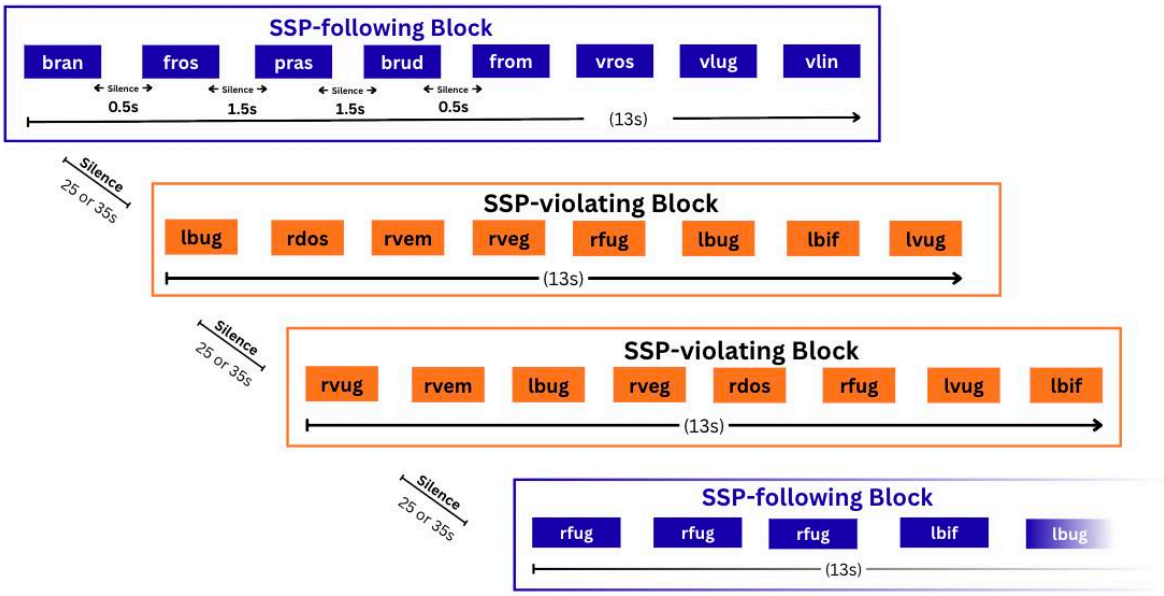
3.4 Experimental design

The study followed a within-groups design, as participants were exposed to both SSP-following and SSP-violating conditions. Therefore, the independent variable was the sonority profile condition, and the dependent variable was the haemodynamic response measured using fNIRS, and specifically, the relative changes in concentrations of Hbo and HbR.

The study used a block design (see Figure 4). The SSP-following or SSP-violating blocks contained the 8 SSP-following or 8 SSP-violating syllables, respectively. Twenty blocks were presented with 10 SSP-following blocks and 10 SSP-violating blocks. The 20 blocks were presented in an interleaved fashion with no more than two consecutive blocks from the same condition. The order of the blocks as well as the syllable order within each block were randomised and counterbalanced across subjects. Within blocks, syllables were separated by silences of either 0.5 or 1.5s, yielding blocks of about 13s long. Blocks were also spaced at time intervals of either 25 or 35s. Both intertrial and inter-block silences were varied in duration to avoid inducing phase-locked brain responses and reduce anticipatory effects that would result from neural entrainment at fixed intervals. The experimental session lasted approximately 12 to 15 minutes.

Figure 4

Schematic illustration of the experimental design used.



3.5 Procedure

Newborns were tested while lying in their cribs in a silent room of the maternity ward of the Hospital San Joan de Déu in Barcelona (Spain). Most participants were tested in a state of quiet sleep.

Parents were provided with the information sheet and signed consent forms before the start of the experimental session. Once signed consent was obtained, the newborn was capped and testing commenced. A computer operated the fNIRS machine and the auditory stimuli were presented through two loudspeakers located on top of a portable hospital cart elevated at the height of the crib, at approximately 30 cm in front of the newborn. The total duration of the testing session lasted 15 to 20 minutes. At the end of the session, a diploma for the newborn was given to parent(s) as an appreciation token. Procedures and protocols were in accordance with the ethical standards of the Helsinki Declaration and approved by the Hospital Sant Joan de Déu Clinical Ethics Committee.

3.6 Data Analysis

3.6.1 Data preprocessing

Data preprocessing and analysis were carried out in Python implementing the Cedalion toolbox (Middell et al., 2026) and using custom Python scripts (García-Castro et al., in preparation). For the full preprocessing pipeline see Appendix B.

For preprocessing, raw light intensity measurements were converted to optical density signal which was subsequently assessed to identify poor-quality channels and motion artifacts. Signal quality was assessed using the Scalp Coupling Index and Peak Spectral Power (SCI $\geq$ 0.60; PSP > 0.04; Beaton et al., 2025) and channels with less than 80% clean recording were removed. Motion artifacts were corrected using Temporal Derivative Distribution Repair (TDDR; Beaton et al., 2025) and wavelet filtering (Molavi & Dumont, 2012). Optical density was then converted into HbO and HbR concentration changes using the modified Beer-Lambert law and the final haemodynamic responses were band-pass filtered between 0.01 Hz and 0.5 Hz to remove low-frequency drift and high-frequency noise. At each event in both conditions, the haemodynamic response signal was segmented into epochs from 5 s before stimulus onset to 28 s after stimulus onset. Within each epoch, motion artifacts were identified

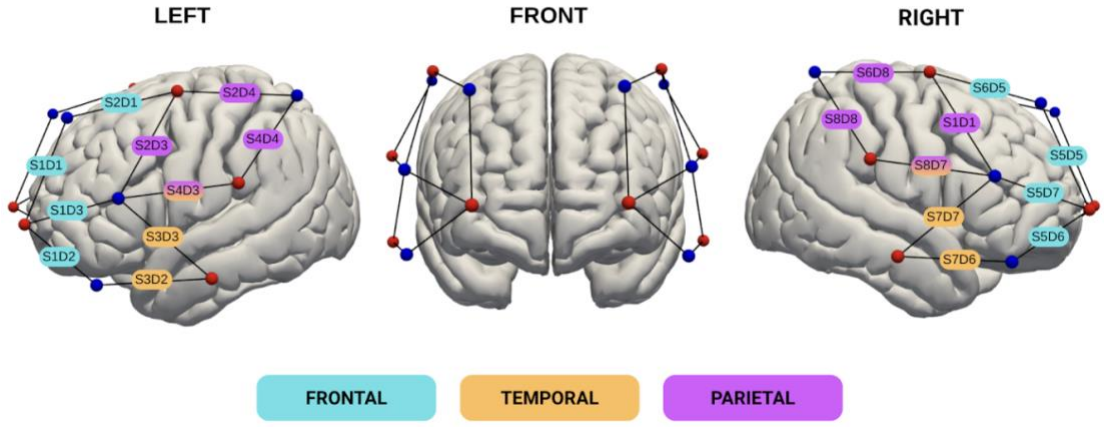
and epochs containing motion artefacts in more than 12 channels were excluded from further analyses. The remaining epochs were baseline-corrected and linear detrending was applied to ensure the beginning and end of each epoch were aligned to the same baseline level. For each participant, the remaining epochs were averaged separately for the SSP-following and SSP-violating conditions to obtain individual haemodynamic response functions (HRFs).

3.6.2 Statistical Analysis

Channels were grouped into six regions of interest (ROIs) following the data-driven classification proposed by Abboub et al. (2016; see Figure 5). Using a localisation analysis in an age-appropriate atlas (Lloyd-Fox et al., 2014), they estimated the cortical regions most likely sampled by each channel and grouped channels according to their anatomical location. This resulted in six ROIs that are also employed in the current study: left frontal, left temporal, left parietal, right frontal, right temporal, and right parietal.

Figure 5

Classification of fNIRS channels into six anatomically defined regions of interest (ROIs) based on the data-driven localisation analysis of Abboub et al. (2016).



For each participant, haemodynamic responses were averaged across all channels within each ROI. The resulting ROI-specific haemodynamic response functions were then averaged across the entire epoch separately for the SSP-following and SSP-violating conditions. This yielded a single mean HbO value and a single mean HbR value for each participant, condition, and ROI.

To assess the research question of whether there was discrimination between conditions, difference scores were calculated subtracting the mean response to SSP-violating stimuli from the mean response to SSP-following stimuli (Following – Violating). One sample *t*-tests were conducted separately for each ROI and chromophore to test whether the mean difference between conditions differed significantly from zero across participants. Importantly, to control for multiple comparisons across the six ROIs, *p*-values were corrected using the Benjamini-Hochberg false discovery rate (FDR; Benjamini & Hochberg, 1995).

4 Results

Results were obtained from 33 participants that were retained for the final analysis. Out of 48 participants, 15 were excluded due to either unusable recordings or no data (n = 8), having more than 10 channels removed (n = 5), and insufficient number of epochs (< 3 epochs per conditions).

Results showed that across all ROIs, relative changes in HbO concentrations were greater for well-formed syllables (SSP-following) than ill-formed (SSP-violating). Concentration difference was calculated as SSP-following – SSP-violating and t-statistics were positive in all regions (see Table 2). After FDR correction, significant differences in HbO concentration between SSP-following and SSP-violating conditions were observed in the left frontal ROI (t = 3.31, p_{FDR} = .008), left temporal ROI (t = 2.88, p_{FDR} = .014), and right temporal ROI (t = 3.26, p_{FDR} = .008).

Table 2

Results of one-sample t-tests comparing HbO responses between SSP-following and SSP-violating conditions across regions of interest (ROIs).

ROI	*t*-statistic	*p*-value	*p*FDR
Frontal Left	3.31	.002	**.008**
Frontal Right	2.16	.039	.058
Parietal Left	2.02	.053	.063
Parietal Right	1.85	.073	.073
Temporal Left	2.88	.007	**.014**
Temporal Right	3.26	.003	**.008**

Note. *p*FDR = Benjamini–Hochberg false discovery rate corrected p-value.
Significant effects after FDR correction (*p*FDR < .05) are shown in bold.

No significant HbR differences between well-formed (SSP-following) and ill-formed (SSP-violating) syllables were observed in any ROI (Table 3). HbR effects were negative, suggesting lower HbR concentration changes in SSP-following compared to SSP-violating conditions. However, none reached statistical significance before or after FDR correction.

Table 3

Results of one-sample t-tests comparing HbR responses between SSP-following and SSP-violating conditions across regions of interest (ROIs).

ROI	*t*-statistic	*p*-value	*p*FDR
Frontal Left	-1.38	.178	.373
Frontal Right	-1.17	.249	.373
Parietal Left	-1.28	.209	.373
Parietal Right	-0.98	.336	.404
Temporal Left	-0.73	.473	.473
Temporal Right	-1.63	.114	.373

Note. *p*FDR = Benjamini–Hochberg false discovery rate corrected *p*-value.

Figure 6 and Figure 7 display the mean HbO and HbR concentration differences (Following − Violating) for the left and right hemisphere ROIs, respectively. Positive values indicate greater responses to SSP-following than SSP-violating stimuli.

As illustrated, HbO responses were positive across all ROIs, indicating greater concentration difference for SSP-following relative to SSP-violating stimuli. The largest condition differences were observed in the left frontal, left temporal, and right temporal regions. In contrast, HbR differences were smaller in magnitude and remained close to zero across ROIs.

Figure 6

Mean HbO and HbR concentration differences (Following − Violating) for left hemisphere ROIs. Error bars represent ± SEM. Significant FDR-corrected effects are indicated by an asterisk.

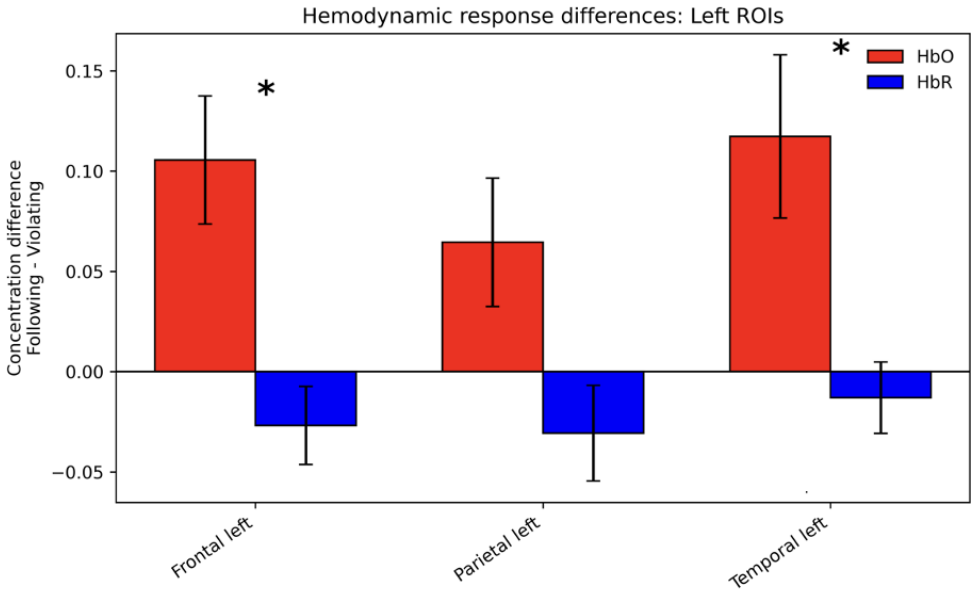
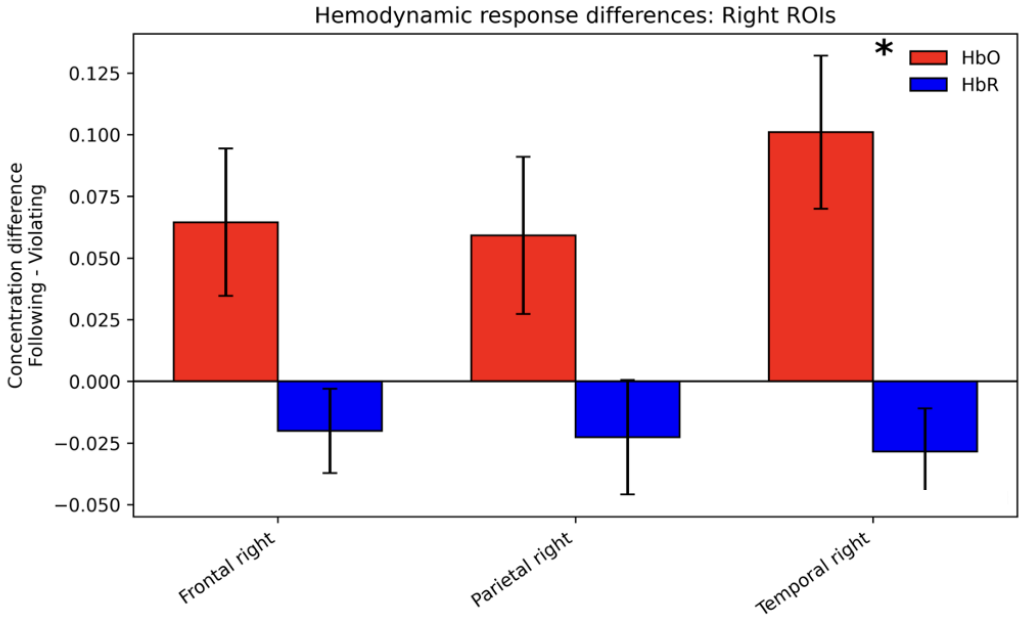


Figure 7

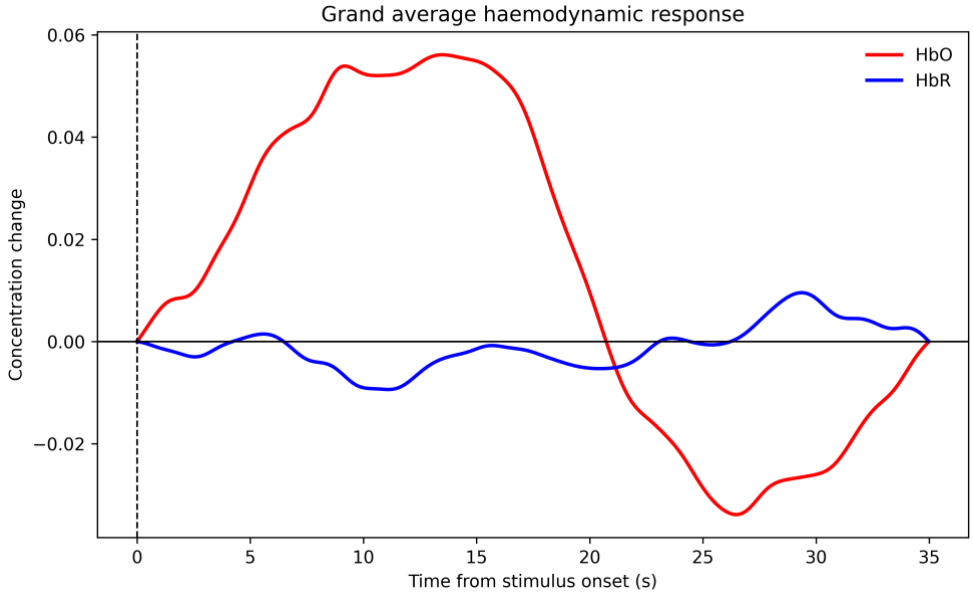
Mean HbO and HbR concentration differences (Following − Violating) for right hemisphere ROIs. Error bars represent ± SEM. Significant FDR-corrected effects are indicated by an asterisk.



In addition, although not directly related to the main research question, a grand average haemodynamic response was analysed across all participants, channels, and conditions to provide a visualisation of the overall haemodynamic response (see Figure 8). This reflects a canonical heamodynamic response function as HbO concentrations increased following stimulus onset, reaching a peak at around 10-15 s post stimulus, while HbR concentrations showed a decrease relative to baseline.

Figure 8

Grand-average haemodynamic response averaged across all participants, channels, and conditions. The dashed vertical line indicates stimulus onset (0 s).



5 Discussion

The present study aimed to conceptually replicate the findings of Gómez et al. (2014) and investigate whether newborns discriminate between well-formed (SSP-following) and ill-formed (SSP-violating) syllables using fNIRS. As Gómez et al. (2014) remains the only study to demonstrate newborns' sensitivity to the SSP, conceptually replicating these findings was important to establish the robustness and reproducibility of this finding. The current findings were consistent with the hypothesis that newborns are sensitive to sonority constraints in speech. Results showed significantly different HbO concentration changes when newborns listened to well-formed compared to ill-formed syllables in the left frontal, left temporal and right temporal regions. These results suggest that newborns discriminated syllables that follow or violate the SSP, supporting the idea that sensitivity to sonority contraints defining the structure of syllables is operating at birth.

In relation to the original study, the present findings represent a successful conceptual replication of Gómez et al. (2014), as both studies demonstrated significantly different haemodynamic responses to well-formed and ill-formed syllables. Gómez et al. (2014)

reported significant effects in the left inferior ROI, corresponding approximately to the left temporal-perisylvian cortex, and in the right superior ROI, encompassing right frontoparietal cortical regions. In the present study, significant effects were observed in the left frontal, left temporal, and right temporal ROIs.

Although the experimental design and speech stimuli were intentionally kept as similar as possible to the original study, several methodological differences may explain the observed differences in the localisation of significant responses. First, Gómez et al. (2014) used a different orientation of the fNIRS probe layout, which was not fully described in the original publication, making direct anatomical comparisons between studies difficult. Furthermore, whereas Gómez et al. analysed broader temporal-perisylvian and frontoparietal regions, the present study classified channels into six anatomically defined ROIs based on the data-driven localisation analysis reported by Abboub et al. (2016) based on a newborn sample by Lloyd-Fox et al. (2014), providing greater anatomical specificity. Nevertheless, despite these differences in probe configuration and ROI definition, the two studies sampled broadly overlapping frontotemporal cortical regions. In addition, the present study implemented a more updated preprocessing analysis pipeline and included a larger sample of newborns, providing greater statistical power than the original study. Together, these methodological differences are likely to account for the differences in the localisation of significant haemodynamic responses between the two studies. Nevertheless, despite these differences, the consistent finding of the involvement of the left temporal regions across both studies suggests that this region may play an important role in the early processing of sonority-based regularities, while also suggesting an additional region, the right temporal region, that may be involved in this processing.

The grand-average haemodynamic response displayed the expected canonical profile, characterised by an increase in HbO and a corresponding decrease in HbR following stimulus onset (Issard & Gervain, 2018). This suggests that the recorded signals reflected a physiologically plausible haemodynamic response and is consistent with the canonical HRF. Gómez et al. (2014) reported an inverted haemodynamic response, a phenomenon that has frequently been observed in newborn fNIRS studies and has been preliminarily attributed to factors such as immature neurovascular coupling (Issard & Gervain, 2018; Lloyd-Fox et al., 2010). Therefore, although the physiological mechanisms underlying inverted and canonical haemodynamic responses remain

unclear, the canonical response observed in the present study provides additional confidence in the quality of the recorded data.

More broadly, the present findings are consistent with previous evidence suggesting that newborns are sensitive to syllable-level structure from the first days of life. Previous studies have shown that newborns discriminate speech stimuli differing in the number of syllables suggesting that the syllable constitutes a perceptually salient unit early in development (Bijeljac-Babic et al., 1993). This is further supported by neurophysiological evidence demonstrating that the newborns preferentially chunk continuous speech into syllable-sized units (Flo et al., 2022), and also evidence supporting that mechanisms underlying syllable discrimination are already functional in preterm infants from approximately 28 weeks of gestation (Mahmoudzadeh et al., 2012). The present findings further support the notion that the syllable is a perceptually salient unit from birth.

Nevertheless, it remains unclear whether newborns' sensitivity to the SSP reflects language-specific phonological processing or a more general auditory processing mechanism. Because well-formed and ill-formed syllables according to the SSP differed in both their phonological and acoustic structure, it is not possible to determine whether the different haemodynamic response was driven by a language-specific processing or by a general sensitivity to the acoustic envelope of the syllables. The present study therefore provides an important proof of concept for the subsequent study of the wider project which aims to address this question. Specifically, the second experiment of the project will apply the same experimental paradigm using non-linguistic tone sequences that preserve the acoustic envelope of syllables while removing their linguistic content. This will allow the contribution of acoustic structure to be examined independently of speech and determine whether newborns' sensitivity to the SSP reflects language-specific processing or a more general auditory processing mechanism.

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7 Statement of contribution

Category	Contributors
Conceptual Ideas	Dr. Chiara Santolin
Methodology	Dr. Chiara Santolin, Dr. Gonzalo García-Castro
Software Programming	Dr. Gonzalo García-Castro
Data Collection	Dr. Gonzalo García-Castro, Chara Sofocleous
Data Analysis	Dr. Gonzalo García-Castro (pre-processing pipeline), Chara Sofocleous (data analysis)
Writing & Review	Chara Sofocleous, Dr. Chiara Santolin, Dr. Gonzalo García-Castro
Original Draft Preparation	Chara Sofocleous

8 Appendices

8.1 Appendix A – Pariticipant Linguistic Profile

Languages spoken in the families of participating newborns.

Language	n (%)
Spanish	25 (75.8)
Catalan	21 (63.6)
English	4 (12.1)
French	2 (6.1)
Arabic	1 (3.0)
Urdu	1 (3.0)
Italian	1 (3.0)

Note. Newborns could be born into bilingual or multilingual families, therefore, percentages do not sum to 100%.

8.2 Appendix B - Preprocessing pipeline script

```
"""Functions for preprocessing fNIRS data."""

from collections import defaultdict
from collections.abc import Sequence
from dataclasses import asdict, dataclass
from itertools import chain
import logging
from pathlib import Path

import numpy as np
from pandas import DataFrame
import xarray as xr

from cedalion.dataclasses import Recording
from cedalion import dot, io, units
from cedalion.nirs.cw import int2od, od2conc
from cedalion.sigproc import motion_correct, quality


logger = logging.getLogger("cedalion")
logger.setLevel(logging.ERROR)

STIM_MAPPING: dict[str, list[int]] = {
    "Following": [9],
    "Violating": [10],
    "Exclude": [4, 6, 7],
}

DURATIONS: dict[str, float] = {"Following": 13, "Violating": 13,
"Exclude": 1}


@dataclass(frozen=True)
class NIRSConf:
    """Configuration parameters for preprocessing.

    Notes:
        Defaults are optimized for neonatal populations.

    Args:
        heart_rate (tuple[float, float], optional): Lower and
upper bound of the heart rate in Hz. Defaults to (1.33, 3.17).
        sci_thresh (float, optional): Scalp Coupling Index (SCI)
threshold. Defaults to 0.6.
        psp_thresh (float, optional): Peak Spectral Power (PSP)
threshold. Defaults to 0.075.
```

```
        filter_freq (tuple[float, float]): Lower and upper bounds
of the band-pass filter in Hz. Defaults to 0.01 and 1.0.
        dpf (tuple[float, float]): Differential Pathlength Factor
(DPF) to use for the 760 nm and 850 nm wavelengths. Defaults to
(4.2, 5.3).
        ma_amp (float, optional): Amplitude threshold for motion
arctifact detection. Defaults to 0.3.
        prune_channels (bool, optional): Should channels marked
as bad be pruned? Defaults to True
        min_time_clean (float, optional): Minimum proportion of
clean recording time should a valid channel have? I less, channel
is marked as bad. Defaults to 0.3.
        epoch_times (tuple[float, float]): Minimum and maximum
time to extract in each epoch, relative to the onset in seconds.
Samples before the onset (negative values) are used for baseline
correction. Defaults to (-5.0, 35.0).
        epoch_detrend (bool, optional): Shoud the epoched signal
be detrended so that onset and offset are set at zero? Defautls
to True.
    """

    heart_freq: tuple[float, float] = (1.33, 3.17)
    sci_thresh: float = 0.60
    psp_thresh: float = 0.04
    filter_freq: tuple[float, float] = (0.01, 0.5)
    dpf: tuple[float, float] = (4.2, 5.3)
    ma_amp: float = 0.3
    ma_std: float = 15.0
    prune_channels: bool = True
    min_time_clean: float = 0.75
    epoch_times: tuple[float, float] = (-5.0, 28.0)
    epoch_detrend: bool = True


def find_pause_offsets(stim: DataFrame, pause_code: int = 6) ->
list[int]:
    """Find pause offsets.

    Args:
        stim (DataFrame): Stimuli dataframe.
        pause_code (int, optional): Trigger code assigned to
pauses. Defaults to 6.

    Returns:
        list[int]: Indices of events in stimuli lists that
correspond to pause offsets.
    """
    offsets = []
    for idx, (i, *_, tt) in enumerate(stim.itertuples()):
```

```
        if int(tt) == pause_code and not idx % 2:
            offsets.append(i)

    return offsets


@dataclass(frozen=True)
class InclusionCriteria:
    """Participant inclusion criteria.

    Args:
        gest_weeks (float, optional): Minimum gestational weeks.
Defaults to 37.
        birth_weight (float, optional): Minimum birth weight in
grams. Defaults to 2700.
        apgar_3 (int, optional): Minimum Apgar score after 10
minutes. Defaults to 8.
        max_pruned_channels (int, optional): maximum number of
pruned channels. Defaults to 8.
        min_epochs (int, optional): Minimum valid epochs in each
condition. Defaults to 2.

    """

    gest_weeks: float = 37.0
    birth_weight: float = 2_700.0
    apgar_3: int = 8
    max_pruned_channels: int = 8
    min_epochs: int = 3


def process_events(stim: DataFrame, **kwargs) -> DataFrame:
    """Group events into their corresponding trial types.

    Arguments:
        stim (DataFrame): Stimuli dataframe.
        **kwargs: Additional arguments passed to
`find_pause_offsets`.

    Returns:
        DataFrame: Stimuli dataframe with grouped trigger labels.
    """
    possible_codes = list(chain(*STIM_MAPPING.values()))
    new_stim = stim.copy()
    lookup = {}

    pauses_remove = find_pause_offsets(new_stim, **kwargs)

    for i, *_, tt in new_stim.itertuples():
```

```
            tt = int(tt)
            if i in pauses_remove:
                new_stim.drop(index=i, inplace=True)
            if tt not in possible_codes:
                new_stim.drop(index=i, inplace=True)
            for k, v in STIM_MAPPING.items():
                if tt in v:
                    lookup[str(tt)] = str(k)

    new_stim.cd.rename_events(lookup)
    new_stim.duration = [
        DURATIONS[tt] if tt in DURATIONS else 0.5 for tt in
new_stim.trial_type
    ]

    return new_stim


def detect_motion_arctifacts(
    rec: Recording, ma_amp: float = 0.4, ma_std: float = 15.0
) -> dict[str, list[tuple[float, float]]] | None:
    win = 1.0 * units.s
    ma_mask = quality.id_motion(rec["od"], win, win, ma_std,
ma_amp)
    rec.set_mask("ma", ma_mask)
    ma_mask = ma_mask.where(ma_mask == 0, drop=True)

    if not any(ma_mask.shape):
        return

    wl = ma_mask.wavelength.values[0]
    segments = defaultdict(list, {ch: [] for ch in
ma_mask.channel.values})

    for ch in ma_mask.channel.values:
        d = ma_mask.sel(channel=ch, wavelength=wl)
        arctifacts = quality.mask_to_segments(d)

        for a in arctifacts:
            segments[ch].append(tuple([float(ai) for ai in a]))

    return segments


def exclude_events(
    df_stim: DataFrame,
    exclude: list[str] | None = None,
    onset: float = -5.0,
    offset: float = 35.0,
```

```
    motion_arctifacts: dict[str, list[tuple[float, float]]] |
None = None,
    max_exc_channels: int | None = None,
) -> tuple[DataFrame, list[int]]:
    """Exclude marked events or events that include marked
events.

    This function takes the stimuli dataframe from a recording,
then returns an updated dataframe with removed events, and a list
of the indices of the excluded events. Annotations are marked for
exclusion if their trial type is in the list provided in
`exclude`, or if the onset of an events marked for exclusion is
inside the onset and offset of the events.

    Args:
        df_stim (DataFrame): Stimuli dataframe for an fNIRs
recording.
        exclude (list[str] | None, optional): List of trial type
labels to be marked for exclusion. Defaults to `["Exclude"]`.
        onset (float, optional): Minimum time to extract in each
epoch, relative to the onset in seconds. Defaults to -5.0.
        offset (float, optional): Maximum time to extract in each
epoch, relative to the onset in seconds. Defaults to 35.0.

    Returns:
        tuple[DataFrame, list[int]]: Updated dataframe with only
included events, and indices of excluded events.
    """
    exc_idx = []
    if exclude is None:
        exclude = ["Exclude"]

    for idx, o, *_, tt in df_stim.itertuples():
        if tt in exclude:
            exc_idx.append(idx)
            continue

        times = o + onset, o + offset
        next_events = df_stim[df_stim.onset.between(*times)]

        if any(ne in exclude for ne in next_events.trial_type):
            exc_idx.append(idx)

    if motion_arctifacts is None:
        return df_stim[~df_stim.index.isin(exc_idx)], exc_idx

    for idx, o, *_ in df_stim.itertuples():
        i = 0
```

```
        for _, ma in motion_arctifacts.items():
            for t1, t2 in ma:
                stim_onset, stim_offset = o + onset, o + offset

                if t1 <= stim_offset and stim_onset <= t2:
                    i += 1
        if max_exc_channels and i > max_exc_channels:
            exc_idx.append(idx)
            continue

    return df_stim[~df_stim.index.isin(exc_idx)], exc_idx


def eval_signal_quality(
    rec: Recording, nirs_conf: NIRSConf = NIRSConf()
) -> tuple[Recording, xr.DataArray]:
    """Evaluate signal quality for each channel.

    This function looks first for channel saturation along the
samples, then calculates the SCI using a 5 second window length,
performs automatic motion arctifact identification, and finally
calculates the proportion of clean time for each channel.

    Args:
        rec (Recording): fNIRs recording. Optical density must
have been calculated already.
        nirs_conf (NIRSConf, optional): Configuration parameters
for preprocessing. Defaults to `NIRSConf()`.

    Returns:
        tuple[cdc.Recoding, xr.DataArray]: Updated recording with
quality masks and proportion of clean time per channel.
    """
    fmin, fmax = nirs_conf.heart_freq
    win = 3 * units.s

    # scalp-coupling index (SCI)
    sci, sci_mask = quality.sci(
        rec["od"],
        win,
        nirs_conf.sci_thresh,
        cardiac_fmin=fmin * units.Hz,
        cardiac_fmax=fmax * units.Hz,
    )
    rec.set_mask("sci", sci_mask)

    # peak spectral power (PSP)
    _, psp_mask = quality.psp(
        rec["od"],
```

```
            win,
            nirs_conf.psp_thresh,
            cardiac_fmin=fmin * units.Hz,
            cardiac_fmax=fmax * units.Hz,
        )
        rec.set_mask("psp", psp_mask)

        # prune channels based on SCI, saturation, and motion
arctifacts
        all_masks = sci_mask & psp_mask
        time_clean = all_masks.sum(dim="time") / len(sci.time)

        return rec, time_clean


def baseline_correct(
    x: xr.DataArray, times: Sequence[tuple[float, float]] | None
= None
) -> xr.DataArray:
    """Apply baseline correction by subtraction.

    This function takes a list of time slices, calculates the
mean of the signal across all of them, and subtracts the mean
from the signal.

    Args:
        x (xr.DataArray): Epochs.
        times (Sequence[tuple[float, float]] | None, optional): A
sequence of time slices (onset, offset) to consider for
calculating the baseline. Defaults to None (no baseline
correction is performed).

    Returns:
        xr.DataArray: Baseline-corrected epochs.
    """
    if times is None:
        return x

    baseline = [x.sel(reltime=slice(tmin, tmax)) for tmin, tmax
in times]
    baseline = xr.concat(baseline, dim="t").mean(["reltime",
"t"])

    return x - baseline


def _border_detrend(signal: xr.DataArray, width: int = 1):
    n = len(signal)
    if n == 0:
```

```
        return signal

    # calculate border indices
    allt = np.arange(n)
    start_indices = np.arange(0, min(width, n))
    end_indices = np.arange(max(0, n - width), n)
    normt = np.concatenate((start_indices, end_indices))

    if len(normt) < 2:
        return signal

    normdata = signal[normt]
    pcoef = np.polyfit(normt, normdata, 1)

    return signal - np.polyval(pcoef, allt)


def detrend_epochs(x: xr.DataArray, width: int = 1, dim: str =
"reltime"):
    """
    Applies border detrending to epochs.

    """
    if dim not in x.dims:
        raise ValueError(f"Dimension '{dim}' not found")

    units = x.pint.units
    x = x.pint.dequantify()

    loop_dims = [d for d in x.dims if d != dim]  # all dims but
time
    # stack non-time dims to a single 'combined_dim'
    # simplifies the broadcasting logic for apply_ufunc
    stacked_data = x.stack(combined_dim=loop_dims) if loop_dims
else x

    out_stack = xr.apply_ufunc(
        _border_detrend,
        stacked_data.pint.dequantify(),
        input_core_dims=[[dim]],
        output_core_dims=[[dim]],
        kwargs={"width": width},
        vectorize=True,
    )

    # restore the original dimensions
    out = out_stack.unstack("combined_dim") if loop_dims else
out_stack
    out = out.transpose(*x.dims)
```

```
    out = out.assign_coords({"trial_type": ("epoch",
x.trial_type.to_numpy())})
    out = out.pint.quantify(units)
    return out


def extract_epochs(
    rec: Recording,
    onset: float = -5.0,
    offset: float = 35.0,
    detrend: bool = True,
    **kwargs,
) -> tuple[Recording, list[int], dict[str, int]]:
    """Extract epoch from fNIRS recoding.

    Args:
        rec (Recording): fNIRs recording. Chromophore
concentrations must have been calculated already.
        onset (float, optional): Minimum time to extract in each
epoch, relative to the onset in seconds. Defaults to -5.0.
        offset (float, optional): Maximum time to extract in each
epoch, relative to the onset in seconds. Defaults to 35.0.
        detrend (bool, optional): Shoud the epoched signal be
detrended so that onset and offset are set at zero? Defautls to
True.

    Returns:
        tuple[xr.DataArray, list[str], dict[str, int]]: fNIRs
recording with extracted epoch, index of excluded epoch, and
number of epoch per trial type.
    """
    motion_arctifacts = detect_motion_arctifacts(rec, **kwargs)
    epoch_idx, exc_epochs = exclude_events(
        rec.stim, onset=onset, offset=offset,
motion_arctifacts=motion_arctifacts
    )

    tt = [k for k in STIM_MAPPING.keys() if k != "Exclude"]
    epochs = rec["conc"].cd.to_epochs(
        epoch_idx, tt, before=abs(onset * units.s), after=offset
* units.s
    )
    bl1 = (onset, 0)
    bl2 = (offset - (offset - onset), offset)
    epochs = baseline_correct(epochs, times=[bl1, bl2])

    # remove baseline periods from data
    epochs = epochs.sel(reltime=slice(0, offset - onset))
```

```
    if detrend:
        epochs = detrend_epochs(epochs)

    rec["epochs"] = epochs

    if epochs.trial_type.data.ndim != 1:
        epoch_unique = epochs.trial_type.to_numpy()[:, 0, 0]
        epoch_unique = [e for e in epoch_unique if e !=
"Exclude"]
    else:
        epoch_unique = [e for e in epochs.trial_type.to_numpy()
if e != "Exclude"]
    keys, counts = np.unique(epoch_unique, return_counts=True)
    counts = {str(k): int(v) for k, v in zip(keys, counts)}

    return rec, exc_epochs, counts


def get_bad_channels(rec: Recording, mask_name: str =
"time_clean"):
    """Get bad channels from recording, based on a given mask.

    Args:
        rec (Recording): fNIRs recording. Optical density must
have been calculated already.
        mask_name (str, optional): Name of the mask to use.
Defaults to "time_clean".

    Returns:
        np.ndarray: Array of bad channels.
    """
    channels = rec["od"].channel.data
    _, bad_channels = np.where(rec.get_mask(mask_name))

    return np.unique(channels[bad_channels])


def preprocess_participant(
    data: dict,
    nirs_conf: NIRSConf = NIRSConf(),
    inc_crit: InclusionCriteria = InclusionCriteria(),
) -> tuple[Recording | None, dict[str, int] | None, list[int] |
None, str]:
    """Preprocess participant NIRS data.

    Arguments:
        data (dict): Participant data.
```

```
        nirs_conf (NIRSConf, optional): Preprocessing parameters.
Defaults to `NIRSConf()`.
        inc_crit (InclusionCriteria, optional): Participant
inclusion criteria. Defaults to `InclusionCriteria()`.

    Returns:
        tuple[Recording | None, dict[str, int] | None, list[int]
| None, str]: Preprocessed fNIRS recording, number of valid
epochs per condition, number of excluded epochs, and inclusion
status of the participant.
    """
    pid = data["participant_id"]
    path = Path("data") / f"sub-{pid:03}"

    if not data.get("data_dir") or not path.exists():
        return None, None, None, "no_data"

    for k, v in data.items():
        for crit, val in asdict(inc_crit).items():
            if k == crit and val > v:
                return None, None, None, crit

    fn = list(path.glob("**/**/*.snirf"))[0]
    rec = io.read_snirf(fn)[0]
    rec.stim = process_events(rec.stim)
    rec.head_model = dot.get_standard_headmodel("icbm152")

    # check if amplitude is within a certain range
    rec["amp"] = rec["amp"].pint.dequantify().pint.quantify("V")
    rec["od"] = int2od(rec["amp"])  # intensity to optical
density (OD)

    # assess the scalp coupling index (SCI) of the channels
    _, time_clean = eval_signal_quality(rec, nirs_conf)
    quality_mask = [time_clean >= nirs_conf.min_time_clean]
    rec.set_mask("time_clean", time_clean >=
nirs_conf.min_time_clean)

    if nirs_conf.prune_channels:
        rec["od"], exc_ch = quality.prune_ch(rec["od"],
quality_mask, "all")

        if len(exc_ch) > inc_crit.max_pruned_channels:
            return rec, None, None, "max_pruned_channels"

    # motion arctifact correction
    rec["od"] = motion_correct.wavelet(rec["od"])
    rec["od"] = motion_correct.tddr(rec["od"])
```

```
    # calculate relative oxy- and deoxy-haemoglobin
concentrations
    dpf_coords = {"wavelength": rec["od"].wavelength}
    dpf = xr.DataArray(list(nirs_conf.dpf), dims="wavelength",
coords=dpf_coords)
    rec["conc"] = od2conc(rec["od"], rec.geo3d, dpf)

    # apply band-pass filter to remove physiological arctifacts
    fmin, fmax = nirs_conf.filter_freq
    rec["conc"] = rec["conc"].cd.freq_filter(fmin=fmin *
units.Hz, fmax=fmax * units.Hz)

    # epoch and baseline-correct the signal
    try:
        onset, offset = nirs_conf.epoch_times

        rec, exc_epochs, epoch_counts = extract_epochs(
            rec,
            onset,
            offset,
            nirs_conf.epoch_detrend,
            ma_amp=nirs_conf.ma_amp,
        )
    except ValueError:
        return rec, None, None, "min_epochs"

    n_epochs = {str(k): int(v) for k, v in epoch_counts.items()}

    if not all(c >= inc_crit.min_epochs for c in
epoch_counts.values()):
        return rec, n_epochs, None, "min_epochs"

    # average epochs from the same trial type
    rec["hrf_conc"] =
rec["epochs"].groupby("trial_type").mean("epoch")

    return rec, n_epochs, exc_epochs, "included"
```